Computer Science & Information Technology

Data Structure and Programming

Trees

DPP: 2

Q1 The minimum number of nodes in AVL tree of height 6 is _____.
(Assume that the height of the root node is 1)

Q2 Consider the following statements:
P: An AVL tree is a height-balanced complete binary tree.
Q: A heap is necessarily a complete binary tree.
Which of the following statement(s) is/are CORRECT?
(A) P only
(B) Q only
(C) Both P and Q
(D) Neither P nor Q

Q3 The total number of ways in which a max-heap can be constructed with the keys-7, 6, 1, 4, 5, 2, 3 is _____.

Q4 Consider the following statements:
P: If the root node of a BST is deleted, it can be replaced by inorder predecessor.
Q: If the root node of a BST is deleted, it can be replaced by preorder successor.
Which of the following is/are CORRECT?
(A) P only
(B) Q only
(C) Both P and Q
(D) Neither P nor Q

Q5 Consider the following operations in a BST-
INSERT(23), INSERT(17), INSERT(25), INSERT(4), INSERT(21), INSERT(1), INSERT(7), DELETE(17), DELETE(23).
The post-order traversal of the resultant BST is
(A) 1, 7, 4, 21, 25
(B) 1, 4, 7, 25, 21
(C) 1, 4, 21, 7, 25
(D) None of the above

Q6 Which of the following sequence(s) of array form a heap?
(A) 23, 17, 14, 6, 13, 10, 1, 12, 7, 5
(B) 1, 5, 10, 6, 7, 12, 13, 14, 17, 23
(C) 23, 17, 14, 7, 13, 10, 1, 5, 6, 12
(D) 1, 5, 10, 12, 13, 7, 14, 17, 23, 6

Q7 Consider the following statements:
P: The accepted balanced factor in an AVL tree are –1, 0 and +1.
Q:AVL trees are self-balancing, ensuring a time complexity of O(log n) for search, insertion, and deletion operations.
The number of INCORRECT statements is _____.

Q8 Construct an AVL tree with the following keys:
12, 10, 15, 14, 13, 17, 8
The immediate left child key value of the root node of the AVL tree is _____.

Answer Key

Q1	20~20	Q5	(A, B, C)
Q2	(B)	Q6	(B, C)
Q3	80~80	Q7	0~0
Q4	(A)	Q8	12~12

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Hints & Solutions

Q1 Text Solution:

The minimum number of nodes in an AVL tree of height 'h' is given by-

$n(h) = n(h-1) + n(h-2) + 1$

$n(1) = 1, n(2) = 2, n(3) = 4, n(4) = 7, n(5) = 12,$

$n(6) = 20$

Q2 Text Solution:

P: INCORRECT. An AVL tree is not necessarily a complete binary tree.

Q: CORRECT. A heap is necessarily a complete binary tree.

Q3 Text Solution:

$$T(n) = 1 * \binom{n-1}{k} * T(k) * T(b-k-1)$$

Here $n = 7, k = 3$

$$T(7) = 1 * \binom{6}{3} * T(3) * T(3)$$

Now, $T(3) = 2$

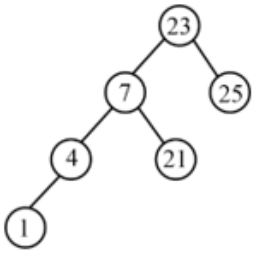
$$T(7) = 1 * \binom{6}{3} * 2 * 2 = 80$$

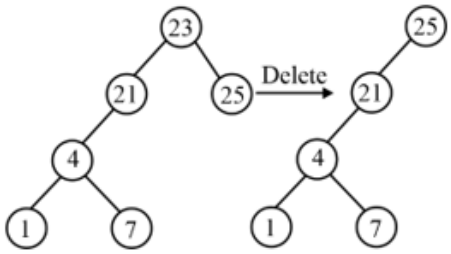
Q4 Text Solution:

If the root node of a BST is deleted, it can be replaced by inorder predecessor/successor.

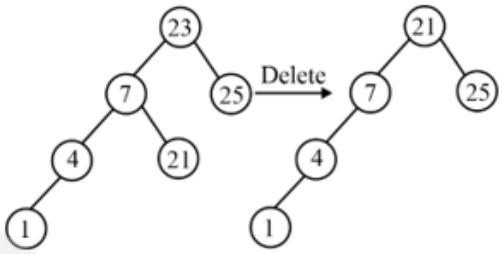
Q5 Text Solution:

or

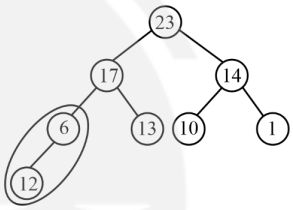




post order traversal: - 1 7 4 21 25

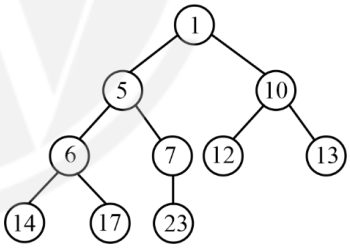


Q6 Text Solution:



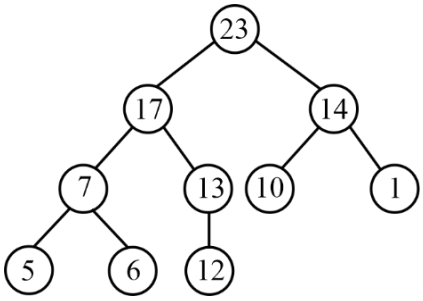
Not possible defies max-hap property

(b)



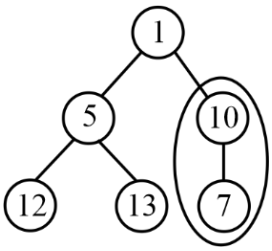
Satisfies min-heap property

(c)



Satisfies max-heap property.

(d)



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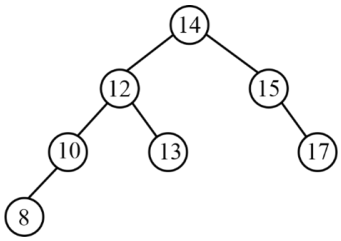
Both the statement are CORRECT.

Q8 Text Solution:

Resultant AVL tree.



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